

# SIRAJUDDEEN MUHAMMADU SHIMRIN

Artificial Intelligence & Machine Learning Engineer

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## PROFESSIONAL SUMMARY

Computer Engineering undergraduate specializing in Artificial Intelligence and Machine Learning with practical experience in computer vision, predictive modelling, and intelligent system development. Skilled in developing ML-based solutions using Python, data analysis techniques, and AI frameworks. Passionate about applying artificial intelligence to solve real-world engineering and healthcare problems.

## EDUCATION

### BSc Engineering (Computer Engineering)

University of Ruhuna, Sri Lanka

2023 – 2027

GPA: 3.43 / 4.00

### Diploma in English Language and Literature

Aquinas College of Higher Studies, Sri Lanka

June 2024

**Relevant Coursework:** Machine Learning, Artificial Intelligence, Data Structures & Algorithms, Computer Vision, Data Analytics.

## TECHNICAL COMPETENCIES

**Programming Languages:** Python, C++, Java, JavaScript

**AI & Machine Learning:** Machine Learning, Scikit-learn, Pandas, NumPy, OpenCV, YOLO, Data Analysis

**Computer Vision:** Object Detection, Image Processing, Video Analytics, Model Evaluation

**Development Tools:** Git, GitHub, Linux, Jupyter Notebook

## AI & MACHINE LEARNING PROJECTS

### Vision-Based Vehicle Speed Estimation | Computer Vision 🔄

- Developed an AI-based traffic monitoring system to detect, track, and estimate vehicle speeds from video streams.
- Applied YOLO object detection, OpenCV image processing, and tracking algorithms for real-time analysis.

### Smart Nutrition Intelligence System (SNIS) | AI Healthcare System 🔄

- Developed an intelligent nutrition recommendation system supporting personalized dietary planning.
- Applied machine learning concepts and data-driven analysis to generate healthcare insights.

### Concrete Strength Prediction | Machine Learning 🔄

- Built a predictive machine learning model to estimate concrete compressive strength using input material parameters.
- Performed data preprocessing, feature analysis, and model evaluation for prediction accuracy.

### Job Searching App | Software Application 🔄

- Developed a recruitment platform demonstrating software engineering practices and user-focused application design.
- Implemented structured data handling and application workflows.

### Bike Race Network | Network Application 🔄

- Developed a digital platform supporting user interaction, event management, and community engagement.
- Applied software architecture concepts for building scalable application workflows.

## EXTRACURRICULARS & COMMUNICATION

**Vice-Captain, Karate Team** | Faculty of Engineering, Ruhuna

**Member** | IESL & IEEE Student Branches

2024 – 2025

Active

**AI Engineering Competencies:**

Machine Learning

Computer Vision

Data Analysis

Model Development

Problem Solving